

The first IRMF component

Notation and main results for smooth absolute loss

At each location, the first IRMF component fits one value to nearby noisy observations. The theory compares that fitted value with an ideal population fit and with the underlying clean signal. This summary omits the proofs.

Symbol	Meaning
u, t	u is an observation location; t is the location where we compute the component. Locations lie on the time axis $[0, 1]$.
n, m	n is the number of observations; m is the number of locations where we compute the component. Often $m = n$.
$X(\cdot), X_u, X_t$	X is the unknown clean signal. $X_u = X(u)$ and $X_t = X(t)$ are its values at the observation and fitting locations.
Y_u	The observed noisy value at location u .
ε_u, σ	$\varepsilon_u = Y_u - X_u$ is observation noise; σ is its standard deviation, in signal-amplitude units.
$h > 0$	Window radius, in time units. Only observations with $ u - t \leq h$ contribute to the fit at t .
$H > 0$	Smoothing scale of the loss, in signal-amplitude units. It controls how closely the loss follows the absolute value.
$a_{u,t}$	The weight of observation u in the fit at t . Weights are nonnegative, fixed before seeing the noise, and sum to one.
$\tilde{S}_t$	The component fitted to the observed data. It changes when the noise realization changes.
S_t	The population component: the minimizer of the loss averaged over repeated noise realizations, for fixed signal, weights, loss scale, and noise law.

The observation model is $Y_u = X_u + \varepsilon_u$, with independent Gaussian errors $\varepsilon_u \sim N(0, \sigma^2)$, meaning mean zero and variance σ^2 . For example, $t = 0.40$ and $h = 0.05$ use observations in $[0.35, 0.45]$.

The loss ρ_H penalizes an amplitude residual r . Its derivative ψ_H , called the score, measures an observation's contribution to the fitting equation:

$$\rho_H(r) = \mathbb{E}_Z |r + HZ|, \quad \psi_H(r) = \rho'_H(r), \quad |\psi_H(r)| \leq 1.$$

Here $Z \sim N(0, 1)$ is an auxiliary standard Gaussian variable, independent of the observation noise. $\mathbb{E}$ means expectation, or averaging over the specified random variable. Primes mean derivatives with respect to the residual. The bounded score limits each observation's contribution.

In the definitions below, z is a candidate fitted amplitude, so $Y_u - z$ is a residual. The notation $\arg \min_z$ means the value of z that minimizes the displayed sum, and $\sum_u$ sums over observations:

$$\tilde{S}_t = \arg \min_z \sum_u a_{u,t} \rho_H(Y_u - z), \quad S_t = \arg \min_z \sum_u a_{u,t} \mathbb{E} \rho_H(Y_u - z).$$

The expectation defining S_t is over observation noise. For Gaussian noise, let B denote the effective loss scale after that averaging:

$$B = \sqrt{H^2 + \sigma^2}, \quad S_t = \arg \min_z \sum_u a_{u,t} \rho_B(X_u - z).$$

Here ρ_B is the same loss with H replaced by B . Thus X_t is the clean signal value, S_t is an ideal local fit, and $\tilde{S}_t$ is the fit computed from noisy data. They generally differ.

Three more quantities describe the estimation error at t :

$F_t > 0$	Population curvature: how strongly the expected loss bends upward near its minimum S_t . It is a scalar measuring local stability.
∇_t	Score fluctuation caused by the observed noise, after subtracting its expectation. Despite the symbol, it is a scalar here.
R_t	The remainder left after subtracting the leading random error ∇_t/F_t from $\tilde{S}_t - S_t$.

Their definitions and the main expansion are

$$F_t = \sum_u a_{u,t} \mathbb{E} \rho_H''(Y_u - S_t) > 0,$$

$$\nabla_t = \sum_u a_{u,t} \{ \psi_H(Y_u - S_t) - \mathbb{E} \psi_H(Y_u - S_t) \}.$$

$$\boxed{\tilde{S}_t - S_t = \frac{\nabla_t}{F_t} + R_t.}$$

Proposition 4.1 bounds R_t . Lemma 4.2 controls the score and curvature fluctuations used in that bound. The corrected value $\tilde{S}_t^{\text{corr}} = \tilde{S}_t - \nabla_t/F_t$ has error R_t relative to S_t . The superscript “corr” labels this correction. It is an *oracle* correction because it uses the unknown S_t and population quantities.

The effective sample size is $n_{\text{eff}}(t) = 1/\sum_u a_{u,t}^2$. For regular weights and enough grid points per window, it is of order h/Δ , where Δ is the grid spacing. On $[0, 1]$, the usual grids have $\Delta = 1/n$ or $1/(n-1)$, both giving order nh . Each individual weight is then at most a constant times $1/(nh)$.

Choose a failure probability $0 < \beta < 1$, and define the logarithmic factor $\ell = \log(4m/\beta)$, using the natural logarithm. For a 95% guarantee, take $\beta = 0.05$. Keep H , σ , and the weight profile fixed; assume the population curvatures F_t stay above a common positive constant. For sufficiently large nh/ℓ , with probability at least $1 - \beta$,

$$\boxed{\begin{aligned} \max_t |\tilde{S}_t - S_t| &\lesssim \sqrt{\frac{\ell}{nh}}, \\ \max_t |R_t| &\lesssim \frac{\ell}{nh}. \end{aligned}}$$

Here $\max_t$ means the largest error across the m fitting locations. The symbol $\lesssim$ means “at most a constant times”; the constants depend on H , σ , the weight profile, and the curvature lower bound. These are upper bounds on the uncorrected and oracle-corrected errors.

The population bias is $S_t - X_t$. If the signal has bounded first and second derivatives and equal weights at equal distances on opposite sides of t , Proposition 4.5 gives $\max_t |S_t - X_t| \lesssim h^2$. Thus h^2 bounds the bias magnitude; it is not the bias itself. Under these conditions, the same probability guarantee gives

$$\max_t |\tilde{S}_t - X_t| \lesssim \underbrace{h^2}_{\text{bias bound}} + \underbrace{\sqrt{\ell/(nh)}}_{\text{random error}} + \underbrace{\ell/(nh)}_{\text{remainder}}.$$

Smaller h tightens the bias bound but raises the random-error bound. Balancing them with h proportional to $(\ell/n)^{1/5}$ gives a bound of order $(\ell/n)^{2/5}$, as $\ell/n \rightarrow 0$.

The results concern the first component on a finite grid with enough local data. Boundary truncation can break symmetry. Asymmetric contamination, later recursive components, and choices of H that change with sample size need additional checks.

See `proposition_4.1_lemma_4.2.tex` for the constants and sample-size conditions, with proofs in Appendix A. This summary uses normalized weights; $x = \log(1/\beta)$ and $z_x = \ell$ in the full note.